

Mini Project 2

ECBME 4090

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Introduction

Write a short description of the experimental paradigm.

The experiment is a motor-imagery EEG paradigm that elicits classifiable patterns of motor cortex activity during imagined left- and right-hand movements. The participant completed multiple runs, each with a sequence of discrete trials. At the beginning of each trial, a visual cue indicated whether the subject should imagine moving their left or right hand. The imagery period lasted several seconds, during which no overt movement was performed; the task relied entirely on imaginary motor simulation.

Continuous EEG was recorded from 16 scalp electrodes, along with auxiliary channels including a trigger channel that marked trial onsets. Each run contained an equal number of left and right imagery trials. For analysis, we isolated the time window corresponding to 4.5 – 8 seconds after trial onset, which captures the sustained motor-imagery phase and excludes visual cue processing and initial transient responses.

This paradigm reliably produces modulations in the mu waves across the sensorimotor cortex. These patterns allow for a CSP filter to maximally separate left- and right-hand imagery, so that the resulting features can be used for LDA classification and evaluate how well motor-imagery intention can be decoded from EEG recordings.

Stimuli and Procedure

Describe your filtering method, sampling frequency, number of experimental blocks and trials, how you separated the two conditions (imagining left vs right), and how you checked the signal quality.

The EEG data was sampled at a frequency of 256 Hz. The filtering method used was CSP filtering, and it works by finding the best linear combination of the maximum differences between patterns of activity across conditions. To remove artifacts and noise, a 0.5 – 30 Hz band-pass filter was applied to the data with a 60 Hz notch filter. There were four experimental blocks (runs) of motor-imagery trials. Each run contained 40 trials each (20 left-hand and 20 right-hand), making 160 trials total and 80 per condition.

A trigger channel marked trial onset, and a separate label channel encoded the cue identity. For each run, the rising edges were detected in the trigger channel to obtain trial start indices, and whether the trial was assigned as “left” or “right” depending on the label code. Signal quality was confirmed by looking at the raw EEG data and ensuring that stable and consistent waves appeared, as well as ensuring that flatlining or signal drift weren’t present. The CSP filters were calculated using $C_L w = \lambda C_R w$, where the first three eigenvectors correspond to filters whose variance is larger for the right-hand condition (small λ) and the last three are filters whose variance is larger for the left-hand condition (large λ). Thus, the first three and last three eigenvectors are the best CSP components to discriminate the two conditions.

Results

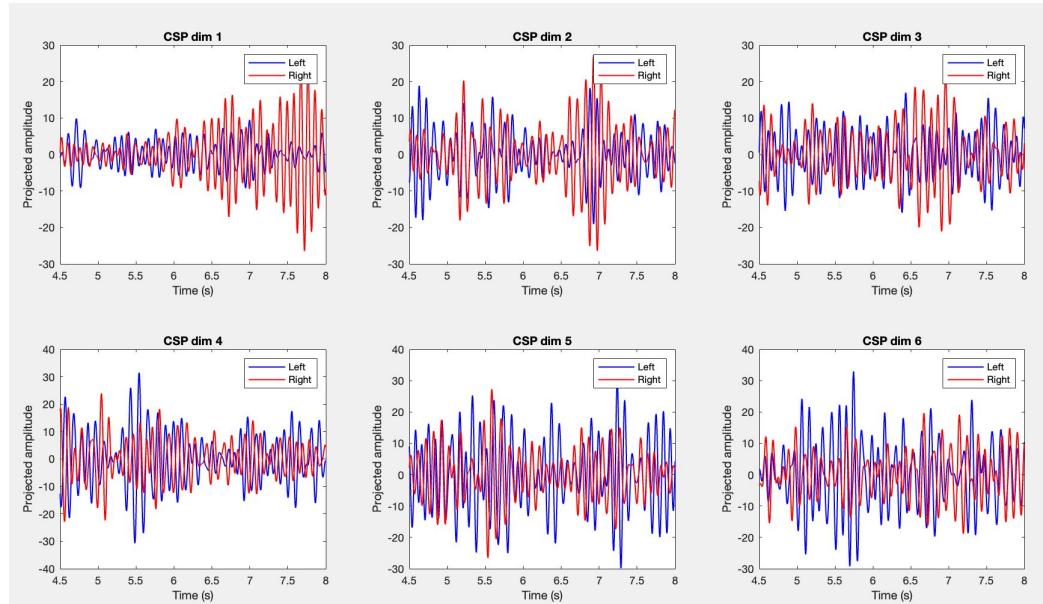
Training Imagined Movement

Describe your behavioral results. Do you see improvement across training sessions? Does accuracy improve over time?

Across the four training sessions, the participant reported an increase ease in doing the tasks and greater consistency when performing the imagined movements. Additionally, there was slight improvement in decoding performance between the two sessions, suggesting that the participant's motor-imagery representations became more consistent with practice as fluctuations within the graph became less frequent. Still, within-session improvements could be made. Classification accuracy in the earlier sessions was generally lower and more variable, reflecting weaker or less stable lateralized sensorimotor patterns. In contrast, later sessions showed higher and more reliable accuracy, indicating that the participant produced more distinguishable left- and right-hand imagery over time. This trend is consistent with typical motor-imagery learning effects, where repeated practice strengthens the corresponding wave modulation. The behavioral results demonstrate a training effect with accuracy improved across sessions, supporting the idea that neural signals and user performance become more separable with continued motor-imagery training.

Applying Common Spatial Pattern Filters

Calculate your CSP filters and explain your steps. Choose the first 3 and last 3 CSP filters corresponding to the largest and lowest Eigenvalues. Plot separately the average projected response for the classes (left vs. right) for each of the projected dimensions.



Applying Common Spatial Pattern Filters

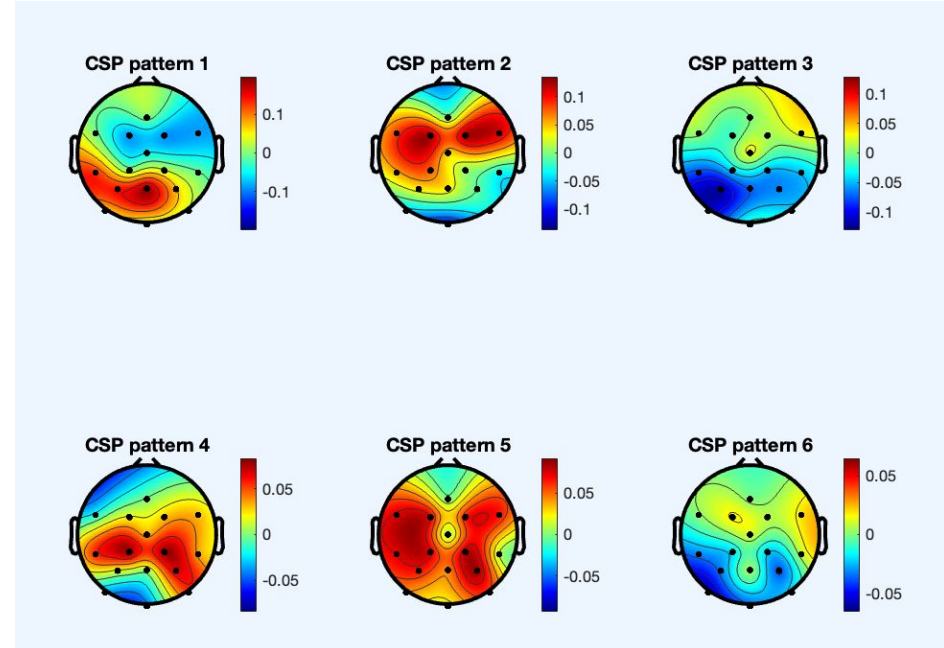
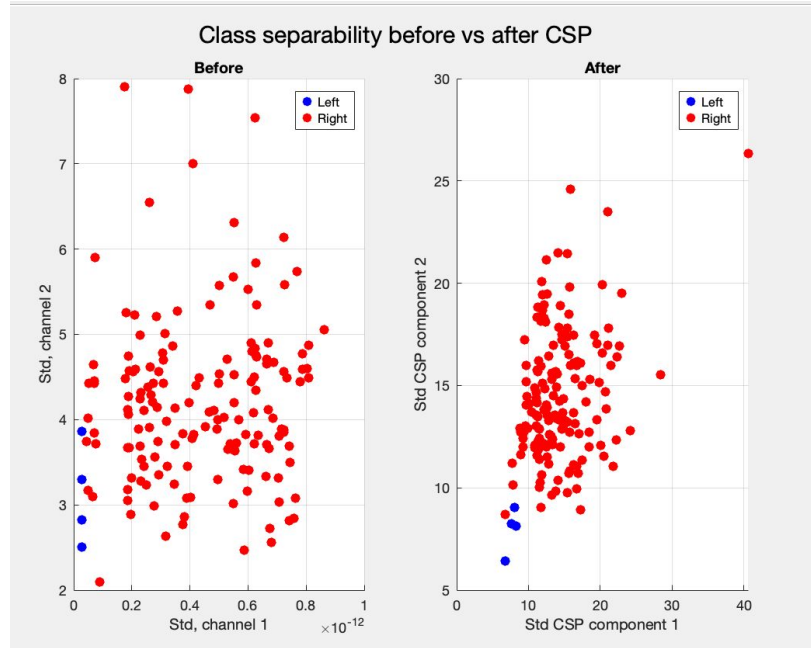
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To compute the CSP filters, I first segmented the band-pass filtered EEG data (7–13 Hz) into left- and right-hand motor imagery trials using the trigger channel for onsets and the label channel to identify each trial as left or right. For each trial, I computed a normalized covariance matrix, where X is the data matrix of size [channels $\times$ time]. I then averaged these trial-wise covariance matrices separately over all left-hand trials to obtain C_L and over all right-hand trials to obtain C_R . The CSP filters were obtained by applying the generalized eigenvalue problem $C_L w = \lambda C_R w$, with MATLAB's $\text{eig}(C_L, C_R)$. This yields eigenvalues λ and eigenvectors w sorted in ascending order. Because small eigenvalues correspond to filters that maximize variance for the right condition relative to left, and large eigenvalues correspond to filters that maximize variance for the left condition, I selected the first 3 and last 3 eigenvectors as my six CSP filters.

To visualize how well these CSP components separate the two classes, I projected each trial into CSP space by computing $Z = W_{\text{CSP}}^T X$, where W_{CSP} contains the six selected filters. This produced time series of length equal to the trial duration for each CSP component and each trial. I then averaged these projected signals across all left-hand trials and separately across all right-hand trials, giving an average time course for each class and each component.

Applying Common Spatial Pattern Filters

Compare the separability of the two classes before and after CSP filtering. What is the effect of spatial filtering on your data? Display the CSP filters on the scalp. Use the standard deviation of the projected data in each trial to remove the time dimension. Explain the benefit of using standard deviation.



Applying Common Spatial Pattern Filters

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Before spatial filtering, the raw EEG signals show substantial overlap between left- and right-hand imagery, with only subtle differences across channels, making the two classes seem difficult to separate. After CSP, the projected components have clearer distinctions as the CSP concentrates class-specific variance into a small number of dimensions, resulting in components that show one condition with consistently higher variance than the other. The corresponding scalp maps show how CSP reweights the electrodes to emphasize lateralized regions that differ between the two tasks. To quantify separability across trials, the standard deviation of each projected component reduced to a single feature was taken, which is appropriate because CSP optimizes variance differences between classes. Using standard deviation captures the information CSP extracts, reduces noise, and provides a clean, interpretable measure of how well each component discriminates left vs. right imagery.

Classification in CSP-Projected Space

Divide your trials into 90% train data and 10% test data. Use your training set to train a Linear Discriminant Analysis classifier and use the Linear Discriminant Analysis classifier to determine the imagined movement direction for your test set. What percentage of trials are correctly classified?

Percentage of trials correctly classified: $78.75\% \pm 0.83\%$ (SE)

Classification in CSP-Projected Space

Repeat this operation ten times with different random train-test splits (ensure they do not overlap). Estimate the standard error of your classification (Standard Error = Standard Deviation / $\sqrt{\text{number of tests}}$).

- Run 1: 56.25%
- Run 2: 62.50%
- Run 3: 71.88%
- Run 4: 59.38%
- Run 5: 59.38%
- Run 6: 75.00%
- Run 7: 59.38%
- Run 8: 71.88%
- Run 9: 75.00%
- Run 10: 59.38%
- **Standard Error of Classification = 0.0264**